

Aditya Bidappa M V

cybersecurity researcher · full-stack security builder · zero-trust systems · ML for defensive security

adityabidappa@gmail.com | github.com/BrownJesús69 | linkedin.com/in/aditya-bidappa | +91 86182 20160 | adityabidappa.netlify.app

B.Tech Cybersecurity · Dayananda Sagar University

CONTACT

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SKILLS

LANGUAGES

Python C SQL Bash PowerShell
TypeScript HTML LaTeX

FRAMEWORKS

FastAPI Express React
React Native Expo
Telegram Bot API

DATA

PostgreSQL MySQL JSONB MongoDB
Qdrant BCNF Schema Design
Data Preprocessing

SECURITY & STANDARDS

Zero Trust NIST SP 800-207
CISA ZTMM v2 CSA SMB Guidance
OWASP Testing Guide v4
IDOR & Auth-Bypass Testing
Blind & Error-Based SQLi
Risk-Aware PKI Authentication
Security Maturity Assessment
IoT Security
DDoS & Attack Classification
Anomaly Detection

PENTESTING TOOLKIT

Burp Suite CE OWASP ZAP
Hoppscotch gobuster nuclei
subfinder nmap Wireshark
SpiderFoot Autopsy Kali Linux
Parrot OS Termux

MACHINE LEARNING

XGBoost scikit-learn
Random Forest Isolation Forest
Multi-Transformer Ensembles
Model Calibration
Model Evaluation Explainable AI
Preregistered Experiments

AUTOMATION & REPORTING

PROFESSIONAL SUMMARY

I'm a third-year Cybersecurity undergrad who treats research and engineering as the same discipline. My work moves between reading standards (NIST SP 800-207, CISA ZTMM v2) and turning them into things that run — CLI tools, deployed web apps, mobile auditors.

The through-line is measurable trust: scoring how exposed a network is, grading an organisation's zero-trust maturity, quantifying how likely a prompt is to be an injection. I care about correctness over shine, and about being honest when a metric is too good to be true.

PROJECTS

PromptGuard

FastAPI · React · TypeScript · XGBoost · scikit-learn

Prompt-injection risk scorer for LLM apps

- ▶ FastAPI risk-scoring service with a React + TypeScript frontend and an XGBoost / scikit-learn backend.
- ▶ Trained on Hugging Face prompt-injection data plus synthetic adversarial prompts generated locally with DeepSeek-R1 1.5B on Ollama.

87.08% accuracy 85% injection recall 95% precision

<https://github.com/BrownJesús69/promptguard>

SME-ZT CLI

Python CLI · CISA ZTMM v2 · NIST SP 800-207 · CSA SMB

Zero-trust maturity assessment for small & medium enterprises

- ▶ A five-pillar, five-level maturity model aligned to CISA ZTMM v2, NIST SP 800-207 and CSA SMB guidance.
- ▶ Weighted scoring with Level-1 capping logic to stop over-scoring, plus detailed textual self-assessment reports.
- ▶ Validated against three documented SME breach case studies and folded into the published manuscript.

<https://github.com/BrownJesús69/zt-maturity-sme>

MUN Argument Builder LIVE

FastAPI · Groq API · Llama 3.1 70B · Render

Deployed argument engine for Model UN delegates

- ▶ Generates structured arguments, counter-arguments, evidence prompts and fallacy checks on demand.
- ▶ FastAPI backend on Groq's Llama 3.1 70B, with GitHub-triggered auto-deploy on Render.
- ▶ Custom domain via GoDaddy and TLS via Let's Encrypt.

<https://quantumpandits.com> <https://github.com/BrownJesús69/mun-argument-builder>

NetworkAudit

React Native · Expo SDK 54 · TypeScript · Express · Zod · Neon PostgreSQL

Real-time Wi-Fi & hotspot security auditor for Android

- ▶ Scans the connected network live and computes a 0–100 security score with an A–F grade.
- ▶ Flags CVE-based findings: default gateway IPs, single DNS server, external-DNS logging risk and SSID identity leakage.
- ▶ Full-stack: RN + Expo client, Express + Zod API, Neon PostgreSQL, EAS Build on Render.

<https://github.com/BrownJesús69/networkaudit>

NutriLog

PostgreSQL · BCNF schema · Frontend · Docs

Relational dietary-logging system

- ▶ Owned the frontend, documentation and application logic for a PostgreSQL nutrition tracker.
- ▶ Correctness-oriented design grounded in a BCNF-normalised schema.

<https://github.com/sampratigaurav/nutrilog>

openpyxl Google Apps Script
GitHub Actions Typst Overleaf

TOOLING

Git GitHub

AI – LLM ASSISTANTS

Claude Claude Code ChatGPT
Gemini DeepSeek Perplexity

AI – VOICE / AUDIO

Whisper

AI – AGENTIC / AUTOMATION

n8n OpenRouter Groq
Claude in Chrome

AI – SECURITY RESEARCH

HexStrike AI
MCP Server Orchestration
LLM-Driven Threat Simulation

AI – AGENT FRAMEWORKS

Z.AI Hermes AI OpenClaw

EDUCATION

B.Tech in Cybersecurity

Dayananda Sagar University, Bengaluru
Aug 2024 – Jun 2028 (Expected)
CGPA 7.5 / 10.0

Coursework: Introduction to Cybersecurity, AI in Cybersecurity, DBMS, Computer Networks

CERTIFICATIONS

Introduction to Cybersecurity

Cisco Networking Academy · Feb 2026

Ethical Hacker

Cisco Networking Academy · Apr 2026

ROLES & ACTIVITIES

President

Cybersecurity Club (2026–27)

Joint Treasurer

Cybersecurity Club (2025–26)

President

DSU MUN Society

Special Advisor

Debate Society

Secretary General

DSU Intra MUN 2026

Secretary General

TNT MUN

Director General

COPE II

Rapporteur

DSUMUN II

Risk-Aware PKI Zero Trust

AI-driven, risk-aware PKI authentication for Zero Trust cloud access

Python · Zero Trust · PKI · Calibrated ML

- ▶ Research prototype that outputs calibrated Low / Medium / High authentication-risk probabilities from certificate-validation evidence and causal behavioural history.
- ▶ Trained and evaluated on nine public security datasets.

Telegram Food Agent

Python · aiogram · Docker · GitHub Actions · Telegram Mini App · Leaflet

Telegram food-ordering bot where an LLM interprets requests and deterministic code enforces the rules

- ▶ Natural-language ordering with an address book, reorder and split flows, photo input, and a Telegram Mini App map picker hosted on GitHub Pages with no backend.
- ▶ Primary model with a fallback; evaluated on dev, held-out and red-team sets, with held-out full-pipeline accuracy rising from 91.7% to 93.8% and zero red-team failures.
- ▶ Released as v0.5.0: ruff, mypy, an 85% coverage gate, Trivy scanning and GHCR image publishing in CI.

93.8% held-out accuracy 438 tests 90% coverage

<https://github.com/BrownJesus69/telegram-food-agent>

EXPERIENCE

Cybersecurity Intern

Jun 2026 – Aug 2026

TabNet Security – Arova Fitness

- ▶ Conducted authorized pentest on arovafitness.com and REST API (api.arovafitness.com) using Kali Linux and Burp Suite Community Edition – DNS/WHOIS recon, TLS review, HTTP header analysis, gobuster directory enumeration, and JavaScript bundle reverse-engineering to uncover an Express.js backend behind Cloudflare and map CORS preflight exposure on authentication-gated endpoints.
- ▶ Parsed OpenAPI specification to catalog 86+ production endpoints across External API, Gym Panel API, and Fitness API services (73 JWT-authenticated, 18 unauthenticated, 22 AI-backed); produced a 352-route API surface map and structured findings reports covering open ports, service versions, and endpoint-level vulnerabilities.
- ▶ Completed two assessment phases yielding 16 total findings (5 Phase 1 unauthenticated, 11 Phase 2 authenticated), including two Critical-severity vulnerabilities – premium plan escalation via PUT / api/user/plan and OTP invalidation flaws – validated via IDOR-style token-swapping and Hoppscotch functional test cases.
- ▶ Architected 192-row exercise library (100 Warmup + 92 Cardio, 12-column schema); built Python/openpyxl data pipeline and Google Apps Script automation to bulk-rename and organize image assets in Drive with lowercase-underscore-numbered convention, validating the platform's live admin category system.

RESEARCH & PUBLICATIONS

Zero Trust Maturity Assessment Framework for Small and Medium Enterprises (SME-ZT)

Co-author

A five-pillar, five-level maturity framework with SME-specific weighting and Level-1 capping to prevent over-scoring, validated across three breach case studies.

Security Threats and Countermeasures in IoMT ML-driven Healthcare Systems

Co-author

Survey of ML-based IoMT intrusion detection, federated learning, privacy preservation and layered defences against malware, DoS and adversarial attacks.

Deepfake Media Analysis: A Survey on Detection Techniques and Open Issues

Co-author

Review of deepfake generation methods, benchmark datasets, detection techniques and the surrounding ethical and legal frameworks.

A Baseline Simulation Framework for DDoS Attack Detection in IoMT Networks

Lead author

Wrote and formatted the full manuscript; showed that 'perfect' metrics on synthetic data arise from trivial one-dimensional separability in the SYN ratio.

AWARDS

Best Delegate: Next Debate MUN'25 · UNSC, Australia
High Commendation: Avent · UNSC, USA
High Commendation: BITS Goa MUN'26 · UNSC, France
Special Mention: MAHE MUN'25 · UNSC, United Kingdom
Verbal Mention: DSBAMUN2 · UNSC, Russia
Honorable Mention: DSU MUN Conference